Saatwik Tiwari

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Professional Summary

Mathematics and Computing student with experience in AI/ML research and software development. Worked on LLM-based applications, machine learning systems, and backend infrastructure through research, internships, and personal projects. Interested in applying AI and software engineering to practical problems.

Education

Birla Institute of Technology and Sciences
B.E. (Hons.) Mathematics and Computing — *Expected May 2028*

- CGPA: 7.64
- Relevant Coursework: Discrete Mathematics, Object Oriented Programming, Data Structures and Algorithms, Linear Algebra and its Applications, Computer Programming, Database Management Systems, Machine Learning

Kautilya Sr. Sec. School — India
Class XII – CBSE — *2023*

- Score: 91.2%

St. Mary Inter College — Etawah, India
Class X – CBSE — *2021*

- Score: 96.2%

Experience

Caarya Innovative — May 2026 – July 2026
Intern — *Remote*

- Built and deployed full-stack **Root Cause Analysis (RCA) agent** using **React, Node.js, and LLM APIs**, automating analysis methods like **Five Whys analysis** with iterative knowledge graphs and chat history; serving **50+ enterprise beta users** with **sub-1s response latency** with configurable LLM settings
- Integrated unit/integration tests and automated CI workflows with code reviews to maintain deployment reliability and code quality. Live Demo

TCS-Research — Jan 2026 – Apr 2026
Research Intern — *Remote*

- Formulated **multi-dimensional evaluation metrics** based on established approaches to quantify **sycophantic behavior in LLMs**.
- Developed a mitigation methodology achieving **25% improvement** over contemporary papers in similar-domain sycophancy-reduction pipelines through **prompt steering and fine-tuning**.
- Authored research manuscript under preparation and planned for submission to relevant conferences and workshops.

Technical Skills

Languages: Python, C++, C, TypeScript, JavaScript, Java, SQL
Frameworks & Backend: React, Node.js, FastAPI, Flask, Express, WebSocket, Prisma
Databases & Storage: PostgreSQL, MongoDB, Zustand
ML/AI Tools: RAG, PyTorch, TensorFlow, JAX, ONNX, LLMs, VectorDB, NumPy, Pandas, Reinforcement Learning
DevOps & Tools: Git, Linux, Railway, AWS, GitHub

Standard Test Scores

JEE Main — Jan/Apr 2024
Percentile: 98.9

BITSAT — May 2024
Score: 298/390

Projects

Crystal – AI Coding Assistant | *Software Engineering, AI Systems*

- Stack: **React** (frontend), **FastAPI + Node.js** (backend), **PostgreSQL**, **ChromaDB** (semantic search)
- Developed end-to-end **AI coding assistant** with real-time repository-wide editing via WebSocket; **indexed 1,000+ files** with incremental embedding pipeline achieving **1,000+ files/min throughput**
- Executed a context-aware **code completion** pipeline, optimized for pro, achieving **sub-2s time-to-first-token latency** through **streaming LLM inference**, batch semantic search with AST parsing, and **RAG assistance**
- Built a FastAPI/Node.js backend combining REST APIs and WebSocket-based real-time communication achieving **low message latency** across **10+ simultaneous users**
- GitHub | Live Demo

OptiMover – Multiplayer Game Platform | *Full-stack Development*

- Stack: **React + TypeScript** (frontend), **Node.js + Express** (backend), **PostgreSQL + Prisma** (data layer), **WebSocket**
- Architected full-stack **multiplayer platform** supporting real-time gameplay between **20+ concurrent players** and multi-level AI agents using **WebSocket** with **Prisma ORM** for persistent session management
- Integrated an Alpha-Zero based **reinforcement learning model** for strategic board games, achieving **52–74% win rate against human players** using Monte Carlo Tree Search with policy and value networks
- Achieved **sub-100ms latency** for **real-time game updates**. Included an Elo-based ranking system using Prisma ORM to manage users, matches, player ratings, and game history
- GitHub | Live Demo

AIMvL – GitHub for ML Models | *Machine Learning, Full-stack Development*

- Stack: **React + TypeScript** (frontend), **FastAPI** (ML backend), Node.js, PostgreSQL
- Implemented a full-stack AI model registry using **React, FastAPI, and Node.js**, supporting **PyTorch/TensorFlow/ONNX** models with model versioning, training, inference, lifecycle management, and deployment through a unified web interface
- Designed **15+ model training recipes** with automated benchmarking across **8+ architectures**, and engineered an asynchronous scheduler for concurrent GPU-accelerated jobs with **Bayesian hyperparameter optimization**, achieving **10–30% performance gains** over baseline configurations
- Implemented an end-to-end reproducible ML pipeline with dataset versioning, artifact storage, and evaluation dashboards, reducing model deployment time from **1 hour to 5 minutes**
- GitHub Repo

AMH-DT – Adaptive Multi-Horizon Decision Transformer | *RL, Finance*

- Constructed a hierarchical encoder discovering latent asset dependency structures via **hypergraph classification**; maps raw multi-asset price data to learned latent embeddings capturing cross-asset temporal dynamics and structural relationships
- Trained a causal policy on K-step trajectories conditioned on return-to-go targets with **L-BFGS turnover penalties**; weight prediction objective delivered a **1.21 Sharpe ratio** and **18% outperformance** relative to standard DT through state-specific decision routing on rolling windows
- Allocated specialized Decision Transformers to each market regime, solving the **non-stationary MDP problem**, avoids mode collapse of single policies by learning regime-specific credit assignment; enables **0.575% monthly alpha** through specialized policy optimization within distinct market behavioral states

Research and Community

- **PRAGYA Research Group:** Active member collaborating on machine learning and systems research
- **SAIDL (Society for AI and Deep Learning):** Member contributing to AI/ML workshops and research endeavors.